

Built-in Template Automation Test Checklist

Scope

Executable checklist for validating implementation against

`docs/specs/built-in-template-automation-test-suite-requirements.md` and `built-in-template-automation-acceptance-criteria.md`.

Pass Standard

- A checklist item is **Pass** only when:
 - command exit code is **0** (unless case expects non-zero),
 - expected output markers are present,
 - required run artifact is persisted.
- For capability-gated scenarios, unmet prerequisites must be marked **Blocked** with reason.

1) Runner Contract Tests

- ☐ `-Mode AllTemplates` executes without parameter ambiguity.
- ☐ `-Mode SelectedTemplates -TemplateIds <single>` executes selected single template only.
- ☐ `-Mode SelectedTemplates -TemplateIds <id1,id2>` executes selected templates only.
- ☐ Unknown template ID returns fail-fast diagnostics and non-zero exit.
- ☐ `-DryRun` prints plan and does not apply templates.

2) Local-Only Guard Tests

- ☐ In local shell (no CI env), runner proceeds normally.
- ☐ With CI indicator set and no override, runner skips/aborts by policy.
- ☐ With CI indicator and explicit override, runner proceeds with warning banner.

3) Metadata Discovery Tests

- ☐ Runner loads template list from `config/templates.json`.
- ☐ All built-in template IDs are included in discovered execution set.
- ☐ Template filtering for selected IDs is deterministic and order-preserving.

4) Environment Preflight Tests

- ☐ Environment snapshot commands execute and are captured:
 - `wsl --status`
 - `wsl --version`
 - `wsl --list --verbose`
- ☐ Target distro exists and can execute basic shell probe.
- ☐ Missing target distro returns clear failure reason.

5) Language Template Probe Tests

5.1 .NET

- ☐ Channel/version validation via `dotnet --list-sdks`.

5.2 Node.js

- ☐ `nvm` availability in new shell session.
- ☐ `node -v` matches selected channel/version family.
- ☐ Selected package manager command is callable.

5.3 Python

- ☐ `pyenv versions` includes selected version.
- ☐ `python --version` matches expectation.
- ☐ Optional Poetry/Pipenv probe succeeds when selected.

5.4 Java

- ☐ `sdk version` callable.
- ☐ `java -version` matches selected channel/version family.

5.5 Rust

- ☐ `rustc --version` callable.
- ☐ `cargo --version` callable.

5.6 Go

- ☐ `go version` matches expected version family.

6) Scenario Template Probe Tests

6.1 Container Runtime

- ☐ Docker Desktop integration mode probe behaves correctly.
- ☐ In-WSL Docker Engine mode probe behaves correctly.
- ☐ Podman mode probe behaves correctly.

6.2 Kubernetes Local

- ☐ Selected cluster path can be created (`kind/k3d/microk8s`).
- ☐ `kubectl get nodes` probe returns expected readiness markers.

6.3 Database Local Stack

- ☐ Selected DB services report running/healthy state.
- ☐ Connectivity smoke probe passes per selected DB.

6.4 AI/ML Profile

- ☐ CPU profile completes and smoke probe passes.

- ☐ GPU profile performs prerequisite checks.
- ☐ Missing GPU prerequisites result in **Blocked** or CPU fallback per policy.

7) Status Classification Tests

- ☐ Successful run is classified as **Pass**.
- ☐ Probe failure with executed checks is classified as **Fail**.
- ☐ Missing host capability is classified as **Blocked**.
- ☐ Final summary counts match per-item status totals.

8) Artifact Persistence Tests

- ☐ XML result file exists and is parseable.
- ☐ JSON manifest exists and includes all executed templates.
- ☐ Markdown summary exists under run-specific docs path.
- ☐ Summary includes pass/fail/blocked counts and environment snapshot.
- ☐ Historical index includes link entry for the run.

9) Reliability and Safety Tests

- ☐ Cancellation path exits gracefully with partial results preserved.
- ☐ Timeout path exits gracefully with clear timeout diagnostics.
- ☐ Logs avoid secrets and sensitive host values.

10) Exit Checklist

- ☐ Full-catalog run completed on at least one prepared local WSL2 distro.
- ☐ Selective single-template run completed.
- ☐ Selective multi-template run completed.
- ☐ No blocker-severity unresolved issues remain for local scope.